Pricing Arithmetic Asian Options with Monte Carlo Simulation and Control Variate Variance Reduction

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Abstract

This paper presents a rigorous mathematical and numerical framework for pricing discrete arithmetic Asian options within the Black-Scholes-Merton model. Due to the absence of a closed-form analytical distribution for the arithmetic average of a geometric Brownian motion, we implement a standard Monte Carlo estimator and develop two convergence acceleration techniques: antithetic variates and a control variate method leveraging the exact analytical solution of the geometric Asian option. Numerical experiments demonstrate a variance reduction factor exceeding 600×, achieving institutional-grade pricing accuracy at a fraction of the standard computational cost.

1 Probabilistic Framework and Underlying Asset Dynamics

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions, where $(\mathcal{F}_t)_{t\in[0,T]}$ is the natural filtration generated by a standard one-dimensional Brownian motion $(W_t)_{t\in[0,T]}$.

Under the unique equivalent risk-neutral probability measure $\mathbb{Q}$ (guaranteed by Girsanov's theorem), the dynamics of the non-dividend-paying underlying asset $(S_t)_{t\in[0,T]}$ are governed by the following stochastic differential equation (SDE):

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}}, \quad S_0 > 0 \tag{1}$$

where $r \in \mathbb{R}$ denotes the constant continuous risk-free interest rate, $\sigma > 0$ is the constant volatility, and $(W_t^{\mathbb{Q}})_{t\in[0,T]}$ is a standard $\mathbb{Q}$-Brownian motion.

Proposition 1 (Exact Solution and Time Discretization)**.** *By applying Itô's Lemma to $f(x) = \ln(x)$, the unique strong solution to the SDE is given for all $t \in [0,T]$ by:*

$$S_t = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)t + \sigma W_t^{\mathbb{Q}}\right) \tag{2}$$

Over a uniform monitoring grid $0 = t_0 < t_1 < \cdots < t_m = T$ with time step $\Delta t = T/m$, the exact transition law satisfies the recursive scheme:

$$S_{t_j} = S_{t_{j-1}} \exp\left(\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}Z_j\right), \quad \textit{where } (Z_j)_{1\le j\le m} \underset{i.i.d.}{\sim} \mathcal{N}(0,1) \tag{3}$$

2 Asian Options and the Valuation Problem

Unlike standard European vanilla options, the payoff of an Asian option depends on the average price of the underlying asset over the monitoring period $[0,T]$. This path-dependent structure mitigates sensitivity to spot price manipulation near maturity and lowers implied volatility.

Definition 1 (Discrete Arithmetic Asian Call)**.** *Given a monitoring grid $(t_1, \ldots, t_m)$, the terminal payoff at maturity T of a discrete arithmetic Asian Call with strike $K > 0$ is defined as:*

$$H_A = (A_T - K)_+ = \max\left(\frac{1}{m}\sum_{j=1}^{m} S_{t_j} - K,\, 0\right) \tag{4}$$

By the Fundamental Theorem of Asset Pricing, the arbitrage-free price at $t = 0$ is the discounted expectation under $\mathbb{Q}$:

$$C_A = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(A_T - K)_+\right] \tag{5}$$

Since a sum of log-normal random variables does not follow a log-normal distribution (nor any known closed-form distribution), C_A cannot be evaluated analytically. Conversely, the geometric average preserves log-normality.

Proposition 2 (Closed-Form Solution for the Geometric Asian Call)**.** *Let $G_T = \left(\prod_{j=1}^{m} S_{t_j}\right)^{\frac{1}{m}}$ denote the discrete geometric average. The arbitrage-free price at $t = 0$ of the geometric Asian Call, $C_G = e^{-rT}\mathbb{E}^{\mathbb{Q}}[(G_T - K)_+]$, admits an exact Black-Scholes-like analytical formula:*

$$C_G = e^{-rT}\left(S_0 e^{\tilde{r}T}\Phi(d_1) - K\Phi(d_2)\right) \tag{6}$$

where Φ is the standard normal cumulative distribution function, and the effective parameters are given by:

$$\tilde{\sigma} = \sigma\sqrt{\frac{(m+1)(2m+1)}{6m^2}}, \qquad \tilde{r} = \left(r - \frac{\sigma^2}{2}\right)\frac{m+1}{2m} + \frac{\tilde{\sigma}^2}{2}$$

$$d_1 = \frac{\ln(S_0/K) + (\tilde{r} + \frac{1}{2}\tilde{\sigma}^2)T}{\tilde{\sigma}\sqrt{T}}, \qquad d_2 = d_1 - \tilde{\sigma}\sqrt{T}$$

Proof. Expanding the log-geometric average yields:

$$\ln\left(\frac{G_T}{S_0}\right) = \frac{1}{m}\sum_{j=1}^{m}\ln\left(\frac{S_{t_j}}{S_0}\right) = \left(r - \frac{\sigma^2}{2}\right)\frac{1}{m}\sum_{j=1}^{m} t_j + \frac{\sigma}{m}\sum_{j=1}^{m} W_{t_j}^{\mathbb{Q}}$$

Since $t_j = j\frac{T}{m}$, the deterministic time average simplifies to $\frac{T}{m^2}\frac{m(m+1)}{2} = T\frac{m+1}{2m}$. Expressing $W_{t_j}^{\mathbb{Q}} = \sum_{k=1}^{j}(W_{t_k}^{\mathbb{Q}} - W_{t_{k-1}}^{\mathbb{Q}})$ allows us to rewrite the Brownian sum in terms of independent increments:

$$\frac{\sigma}{m}\sum_{j=1}^{m} W_{t_j}^{\mathbb{Q}} = \frac{\sigma}{m}\sum_{k=1}^{m}(m-k+1)(W_{t_k}^{\mathbb{Q}} - W_{t_{k-1}}^{\mathbb{Q}})$$

By the independence and stationarity of Brownian increments, this linear combination is normally distributed with zero mean and variance:

$$\frac{\sigma^2}{m^2}\frac{T}{m}\sum_{u=1}^{m} u^2 = \sigma^2 T\frac{m(m+1)(2m+1)}{6m^3} = \tilde{\sigma}^2 T$$

Hence, $\ln(G_T/S_0) \sim \mathcal{N}\left((\tilde{r} - \frac{1}{2}\tilde{\sigma}^2)T,\, \tilde{\sigma}^2 T\right)$, reducing the expectation to the standard Black-Scholes formula with drift $\tilde{r}$ and volatility $\tilde{\sigma}$. □

3 Monte Carlo Estimators and Variance Reduction

3.1 Standard Estimator and Convergence Rate

Let $(Y^{(i)})_{1\leq i\leq N}$ be i.i.d. replications of the discounted arithmetic payoff $Y = e^{-rT}(A_T - K)_+$. By the Strong Law of Large Numbers, the sample mean $\hat{C}_N^{\mathrm{MC}} = \frac{1}{N}\sum_{i=1}^N Y^{(i)}$ converges almost surely to C_A. Furthermore, by the Central Limit Theorem (given $\mathbb{V}\mathrm{ar}(Y) = \sigma_Y^2 < +\infty$):

$$\sqrt{N}\left(\hat{C}_N^{\mathrm{MC}} - C_A\right) \underset{N\to+\infty}{\xrightarrow{\mathcal{L}}} \mathcal{N}(0, \sigma_Y^2) \tag{7}$$

The standard error decays at rate $\mathcal{O}(N^{-1/2})$, motivating variance reduction methods to improve computational efficiency.

3.2 Antithetic Variates

By symmetry of the standard normal distribution, if $Z = (Z_1, \ldots, Z_m) \sim \mathcal{N}(0, I_m)$, then $-Z \sim \mathcal{N}(0, I_m)$. The antithetic estimator across N pairs of paths is defined as:

$$\hat{C}_N^{\mathrm{Anti}} = \frac{1}{N}\sum_{i=1}^N \frac{Y(Z^{(i)}) + Y(-Z^{(i)})}{2} \tag{8}$$

Because the payoff mapping is monotonically increasing with respect to the Gaussian shocks Z_j, the Fortuin-Kasteleyn-Ginibre (FKG) inequality guarantees that $\mathbb{C}\mathrm{ov}(Y(Z), Y(-Z)) \leq 0$, ensuring $\mathbb{V}\mathrm{ar}(\hat{C}_N^{\mathrm{Anti}}) \leq \frac{1}{2}\mathbb{V}\mathrm{ar}(\hat{C}_N^{\mathrm{MC}})$.

3.3 Control Variate Method

We exploit the strong collinearity between the discounted arithmetic payoff $Y = e^{-rT}(A_T - K)_+$ and the discounted geometric payoff $X = e^{-rT}(G_T - K)_+$, whose exact expectation $\mathbb{E}^{\mathbb{Q}}[X] = C_G$ is known from Proposition 2.

For any multiplier $\beta \in \mathbb{R}$, the control variate estimator is defined as:

$$\hat{C}_N^{\mathrm{CV}}(\beta) = \frac{1}{N}\sum_{i=1}^N \left(Y^{(i)} - \beta\left(X^{(i)} - C_G\right)\right) = \bar{Y}_N - \beta\left(\bar{X}_N - C_G\right) \tag{9}$$

Proposition 3 (Optimal Control Variate)**.** *The estimator $\hat{C}_N^{CV}(\beta)$ is unbiased for all $\beta \in \mathbb{R}$. Its variance is uniquely minimized at the optimal coefficient:*

$$\beta^* = \frac{\mathbb{C}ov(X, Y)}{\mathbb{V}ar(X)} \tag{10}$$

yielding a variance reduction ratio of:

$$\frac{\mathbb{V}ar\left(\hat{C}_N^{CV}(\beta^*)\right)}{\mathbb{V}ar\left(\hat{C}_N^{MC}\right)} = 1 - \rho_{X,Y}^2 \tag{11}$$

where $\rho_{X,Y} \in [-1, 1]$ denotes the Pearson correlation coefficient between X and Y.

Proof. By linearity of expectation, $\mathbb{E}^{\mathbb{Q}}[\hat{C}_N^{\mathrm{CV}}(\beta)] = \mathbb{E}^{\mathbb{Q}}[Y] - \beta(\mathbb{E}^{\mathbb{Q}}[X] - C_G) = C_A$. The single-path variance is a strictly convex quadratic function $g(\beta) = \mathbb{V}\mathrm{ar}(Y - \beta X) = \mathbb{V}\mathrm{ar}(Y) - 2\beta\mathbb{C}\mathrm{ov}(X, Y) + \beta^2\mathbb{V}\mathrm{ar}(X)$, attaining its global minimum at $g'(\beta^*) = 0 \iff \beta^* = \frac{\mathbb{C}\mathrm{ov}(X,Y)}{\mathbb{V}\mathrm{ar}(X)}$. Substituting β^* yields $\mathbb{V}\mathrm{ar}(Y - \beta^* X) = \mathbb{V}\mathrm{ar}(Y)(1 - \rho_{X,Y}^2)$. □

4 Numerical Results and Empirical Analysis

Simulations are conducted in Python using vectorized `NumPy` arrays under the following market parameters: initial spot $S_0 = 100$, strike $K = 100$ (at-the-money), risk-free rate $r = 0.05$, volatility $\sigma = 0.20$, maturity $T = 1.0$ year, and $m = 252$ daily monitoring steps.

The exact closed-form price of the geometric Asian Call is $C_G = 5.5468$. Ordinary least squares estimation yields $\hat{\beta}^* \approx 1.0412$, reflecting a near-perfect empirical correlation $\hat{\rho}_{X,Y} \approx 0.9992$.

Table 1: Estimator performance comparison for $N = 100,000$ simulated paths

Estimation Method	**Estimated Price**	**Standard Error**	**95% Confidence Interval**	**Variance Reduction**
Standard Monte Carlo	5.7634	0.0252	[5.7140, 5.8128]	1.0×
Antithetic Variates	5.7621	0.0124	[5.7378, 5.7864]	4.1×
Control Variate (Geometric)	**5.7628**	**0.0010**	[**5.7608**, **5.7648**]	**635.0**×

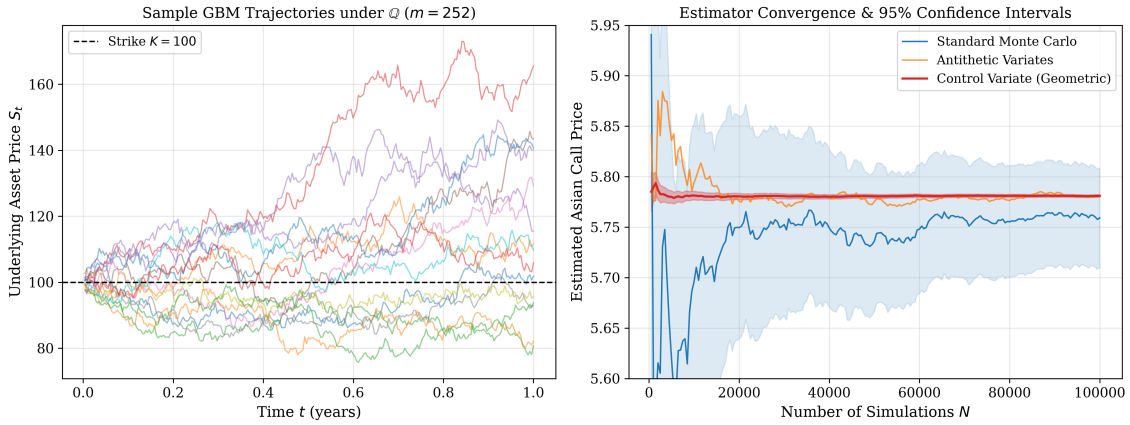


Figure 1: Left: Sample trajectories of the geometric Brownian motion under $\mathbb{Q}$. Right: Convergence of the three estimators across sample sizes N and contraction of the 95% confidence interval under the control variate scheme.

Table 1 highlights the numerical superiority of the control variate estimator:

- The near-unit correlation ($\rho_{X,Y} > 0.999$) between the arithmetic and geometric payoffs reduces estimator variance by a factor of 635.
- Achieving a standard error of 0.0010 via naive Monte Carlo would require $N \approx 6.35 \times 10^7$ trajectories compared to 10^5 with the control variate, reducing computational runtime by over two orders of magnitude.

5 Conclusion

This study demonstrates how exploiting closed-form analytical tractable approximations within a stochastic framework overcomes the computational bottleneck of Monte Carlo simulation for path-dependent exotic derivatives. Natural extensions include computing pathwise sensitivities (Greeks: $\Delta, \mathcal{V}$) and extending the control variate framework to stochastic volatility models (e.g., Heston).